Minimizing the Influence of Cogging Torque on Vibration of PM Brushless Machines by Direct Torque Control

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Direct torque control (DTC) is employed to minimize the influence of the cogging torque on the vibrational behavior of a PM brushless machine. Simulated and measured results, both with and without cogging torque compensation, show that DTC is effective in compensating for the influence of cogging torque provided it is taken into account in the torque estimation. The method results in a simple and effective control algorithm and significantly reduces torque ripple and vibration.

***Index Terms*—Acoustic noise, brushless machine, cogging torque, direct torque control, permanent magnet, vibration.**

I. Introduction

COGGING torque results from the interaction of the rotor permanent magnets with the stator slots. It may cause speed ripple and induce vibrations of coupled components, which radiate acoustic noise [1], particularly under light load and low speed conditions. Many applications, such as electric power-assisted steering and direct-drive servo systems, are extremely sensitive to torque ripple. Thus, various design features may be employed to minimize the cogging torque. These include the use of skew, an optimal magnet pole-arc to pole-pitch ratio, auxiliary slots and teeth, and a fractional number of slots per pole. However, most of these compromise other aspects of the electromagnetic performance, such as the back-emf waveform, and generally increase the cost. Further, although they are effective, they cannot totally eliminate the influence of the cogging torque, while their utility is affected by manufacturing tolerances, etc. An alternative approach to suppressing the influence of cogging torque, if it is relatively small, is to optimize the phase current waveform [2], albeit increasing the complexity of current control.

Direct torque control (DTC), which was originally proposed for 3-phase induction machines, directly controls the stator flux-linkage and the electromagnetic torque, and considers the machine, its power electronic inverter and the control strategy at the system level, in that a direct relationship is established between the electromagnetic torque and flux and the optimal inverter-switching sequence, so as to achieve a fast torque response and good dynamic performance. Recently, it has been successfully applied to 3-phase PM brushless AC [3] and brushless DC drives [4]. To date, however, the influence of cogging torque has not been considered in the implementation of DTC.

This paper applies DTC to a PM brushless machine, which may be either brushless AC or DC, highlights the difference in its implementation and compares the performance both with and without cogging torque compensation, and demonstrates the effectiveness of the proposed method in reducing the influence of cogging torque on the vibration behavior, by both simulation and experiment.

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II. Direct Torque Control Both With and Without Consideration of Cogging Torque

Although the DTC technique is applicable to any type of PM brushless machine [4], both with and without rotor saliency, for simplicity only nonsalient brushless machines equipped with surface-mounted magnet rotors will be considered. For such machines which may have a nonsinusoidal stator flux-linkage waveform, the electromagnetic torque, for both BLAC and BLDC operation, can be simplified as

$$T_e = \frac{3}{2}\frac{p}{2}\left[\frac{d\psi_{r\alpha}}{d\theta_e}i_{s\alpha} + \frac{d\psi_{r\beta}}{d\theta_e}i_{s\beta}\right] \tag{1}$$

where θ_e is the rotor electrical angle, p is the number of poles, and i_{sa}, $i_{s\beta}$, ψ_{ra}, and $\psi_{r\beta}$ are the α- and β-axis stator currents and rotor flux-linkages, respectively.

In the DTC implementation [3], [4], the stator flux-linkages are observed from the measured stator voltages and currents, while the rotor flux-linkages are deduced from the stator flux-linkages. Both the flux-linkage and torque are directly controlled by the optimal inverter switching table, from which the inverter voltage space vectors are determined from the difference between the commanded and actual torque and flux-linkage. In conventional DTC, (1) is used for torque feedback and the cogging torque is neglected. In the proposed DTC, the influence of the cogging torque is accounted for by effectively considering it as an external load, and (2) is used for torque feedback, i.e.,

$$T = T_e + T_c \tag{2}$$

where T_c is the cogging torque, which may be precalculated, either by an analytical method or by finite element analysis, or measured. Since the torque is directly controlled, potential sources of torque ripple, due, for example, to a machine having a nonideal back-emf waveform as well as due to cogging torque, will be eliminated if these are accounted for in the torque estimation. As will be evident from (2), it is relatively easy to compensate for the influence of cogging torque, as long as the bandwidth of the torque controller is sufficiently high.

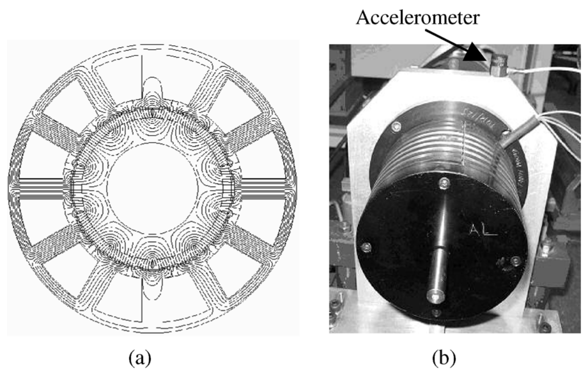


Fig. 1. (a) Open-circuit field distribution and (b) prototype motor with accelerometer placed on mounting frame.

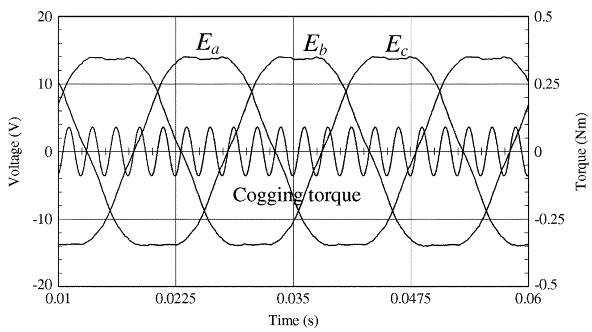


Fig. 2. Measured phase back-emf waveforms and cogging torque waveform.

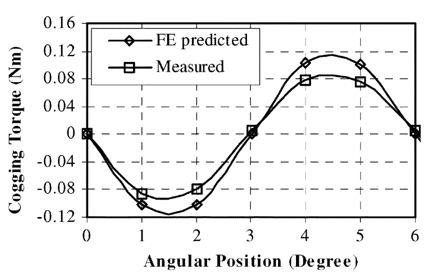


Fig. 3. Finite element predicted and measured cogging torque waveforms.

III. Simulated and Measured Results

A prototype 3-phase, 10-pole, 12-slot, 400 rpm, brushless DC motor, whose design parameters are given in [4], is used to validate the proposed DTC method, which was implemented with a TMS320C31 DSP control system and simulated by Matlab/Simulink based model. Fig. 1 shows the open-circuit field distribution, together with the prototype motor undergoing a vibration test, while Fig. 2 shows measured phase back-emf and cogging torque waveforms. Fig. 3 compares finite element predicted and measured cogging torque waveforms. Since the least common multiple between the number of slots and the number of poles is 60, the periodicity of the cogging torque waveform is 6° mech., while that of the back-emf and current waveforms is 36° mech. Hence, within each back-emf cycle, there are 6 cogging torque cycles. Further, due to the fractional number of slots per pole, the cogging torque waveform is essentially sinusoidal, although DTC can account for any waveform. The cogging torque was calculated by Maxwell stress integration and magnetostatic finite element analysis, and was measured by clamping the stator at various angular positions relative to the rotor and measuring the static torque

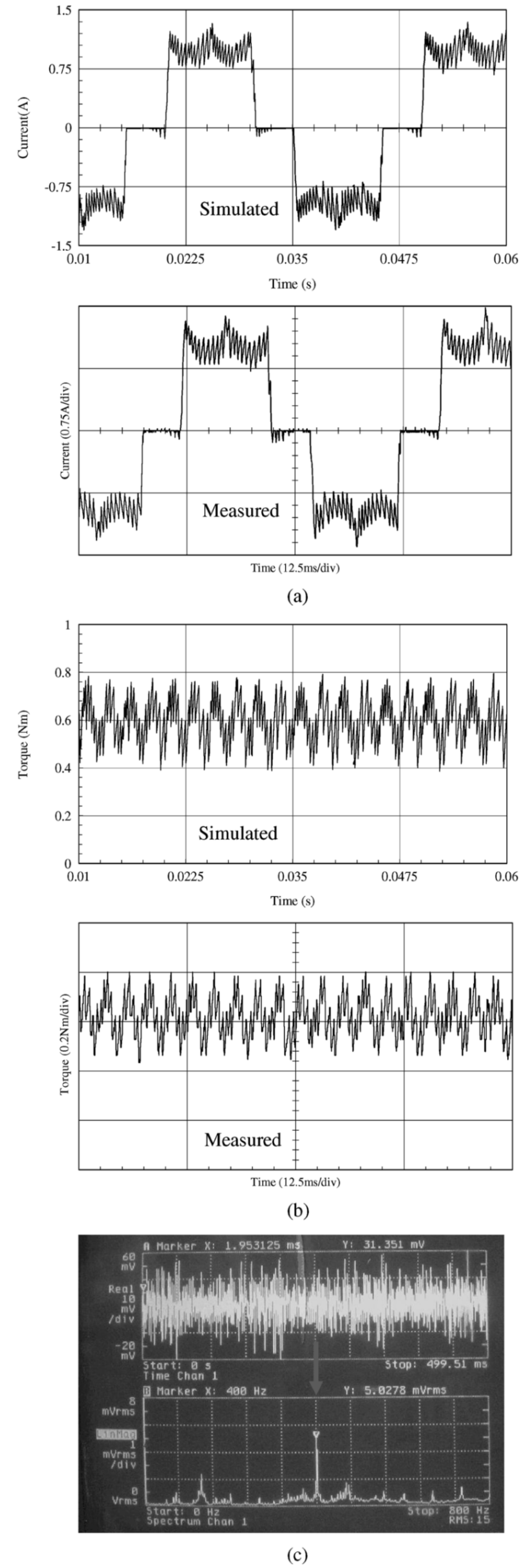
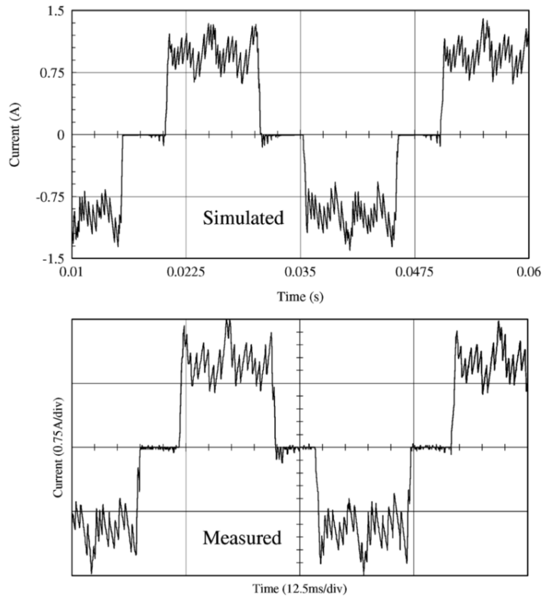
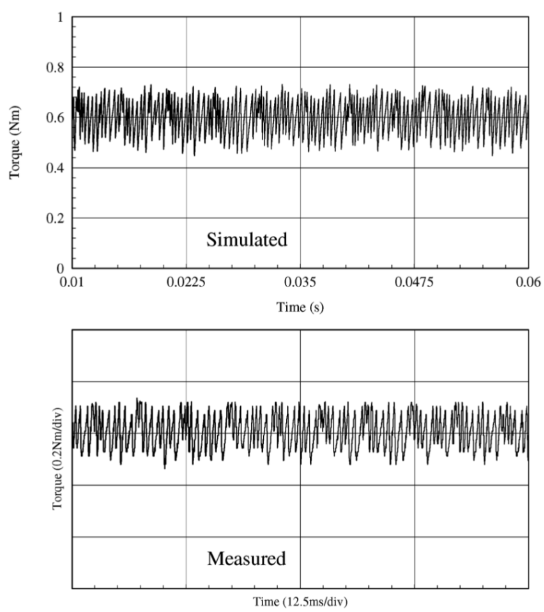


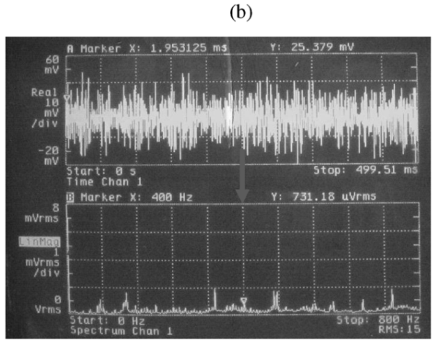
Fig. 4. Measured current and torque waveforms and vibration responses—without cogging torque compensation. (a) Phase current. (b) Torque. (c) Vibration (upper trace: vibration, lower trace: frequency spectrum).



(a)



(b)



(c)

Fig. 5. Measured current and torque waveforms and vibration responses—with cogging torque compensation. (a) Phase current, (b) Torque, (c) Vibration (upper trace: vibration, lower trace: Frequency spectrum).

which acts on the rotor. It can be seen that, probably due to manufacturing tolerances and tolerances in the properties of the sintered NdFeB magnets (for which a remanence of 1.2 T was assumed in the calculation) as well as measurement errors, there is a significant difference in the predicted and measured cogging torque. In the simulation and DSP implementation, therefore, the measured cogging torque waveform was employed. In addition, the measured torque was obtained from the torque observer, i.e., (2).

Fig. 4 compares simulated and measured current and resultant torque waveforms, and includes the measured tangential vibration response and frequency spectrum, which was measured by an accelerometer placed circumferentially on the mounting frame, Fig. 1(b), when the influence of cogging torque is neglected, i.e., the feedback torque was obtained from the torque observer, (1). As will be seen, whilst excellent agreement is achieved, since the feedback torque from (1) is not the actual torque, the resultant torque exhibits a significant torque ripple due not only to the cogging torque but also to a PWM induced torque ripple. A significant 400 Hz frequency component is evident in the tangential vibration due to the influence of the cogging torque, since its frequency is given by $nN_c/60$, where n is the rotor speed and N_c is the least common multiple between the number of slots and the number of poles [5]. Fig. 5 compares simulated and measured current and resultant torque waveforms, and again includes the measured vibration response and frequency spectrum, which result when the cogging torque is accounted for in the torque observer, i.e., (2) is used for torque feedback. As can be seen, the current waveform automatically adapts to compensate for the influence of the cogging torque. As a result, the 400Hz vibration component is reduced significantly, although a high frequency ripple due to PWM is still evident. Thus, the effectiveness of DTC in compensating for the influence of the cogging torque has been demonstrated.

IV. Conclusion

The influence of cogging torque on the vibrational behavior of PM brushless machines can be minimized by DTC. The effectiveness and simplicity of DTC has been validated by comparing simulated and measured current and torque waveforms, as well as the measured vibration, both with and without cogging torque compensation.

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